

CSE2315 — Assignment 7

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Exercise 1

Prove that the following problem is in NP:

$\{(V, E) \mid \text{the graph } G = (V, E) \text{ contains a simple path } \pi \text{ such that } \forall v \in V : v \in \pi\}$

In a different notation we would write:

Name: Hamiltonian Cycle (HAMC)

Input: A graph $G = (V, E)$.

Question: Is there a simple cycle π in G such that all vertices $v \in V$ are in that cycle?

Exercise 2

Old exam question, endterm 23–24:

For the following claim, if it is true: prove it. If it is false, give an explanation showing the claim is false and if possible a counterexample. Remember that the cover of our exam reads: “Unless otherwise stated we assume that $P \neq NP$ and that $NP \neq EXP$.”

For three languages A, B, C if $A \leq_P B$ and $B \leq_m C$ with $C \in NPC$. Furthermore we know that there is a regular expression R such that $L(R) = A$. Then $B \in P$.

Exercise 3

Old exam question:

On the (famous?) British Quiz Show “The University Challenge”, the following question was asked in the grand final of 2018/2019.

... which problem in computing asks whether all problems with a solution verifiable in finite time, can also be solved in finite time.

The contestants answer with “P equals NP” and the host answers back with “Yes that’s right, P versus NP, yes”.

Assuming this is the answer the host was looking for, what is wrong with the question he asks?

Exercise 4

Old exam question, resit 20–21:

(a) Prove that the following problem is in NP.

Name: AmazingCustomChallenge

Instance: Given a set L of lectures, a set of students S , a collection of subsets T with $|T| = |S|$ so that T_i is the set of lectures that student i needs to attend, and an integer K .

Question: Is there a function $f : L \rightarrow \{1, 2, \dots, K\}$ that maps lectures to time slots so that all students can attend all their lectures. In other words, is there a function so that there is no student who has to attend two lectures that are mapped to the same time slot.

(b) Not originally part of the exam question. Your answer to the previous question should have included time complexities. If we are to describe them in terms of $|I|$ where I is an instance of the problem, what would be the complexity? (Hint: first think about $|I|$, is it $|S|$, or $|L|$, or some combination? How does K factor into this?)

Exercise 5

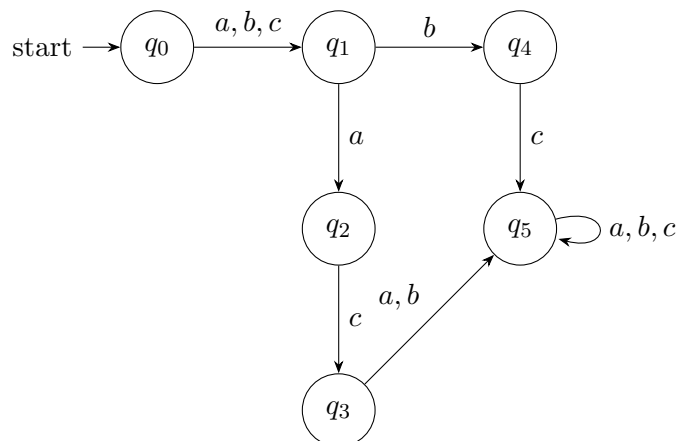
Variation on old exam question, resit 20–21:

Show the claim is true or false, starting your answer with either “True” or “False”.

Given two problems A and B that can be solved using a PDA, it is possible to create a Karp-reduction from A to B and vice versa.

Exercise 6

Revision, old exam question, resit 20–21: Given this NFA N , give a regular expression R so that $L(N) = L(R)$.



Exercise 7

Revision, old exam question, endterm 16–17: Suppose we have the following language:

$$L = \{\langle M, v, w \rangle \mid M \text{ outputs } w \text{ on the tape when run on input } v\}$$

Use a mapping reduction to prove that L is undecidable. Give:

- (a) A suitable problem to reduce from.
- (b) A suitable reduction function $f : \Sigma^* \rightarrow \Sigma^*$.
- (c) A proof showing f satisfies the requirements of a mapping reduction.

Bonus Exercise